

RememberBirthdays

A full-stack Spring Boot application to help users remember and manage birthdays, featuring secure authentication, automated email reminders, and a modern React frontend deployed on AWS.

Problem

Most people rely on social media to track friends' birthdays. RememberBirthdays stores birthdays securely in a private, user-scoped database and sends automatic daily email reminders -- no social platform required.

Tech Stack

Category	Technology
Language	Java 21
Framework	Spring Boot 3.5
Security	Spring Security, OAuth2 Resource Server, Keycloak
Persistence	Spring Data JPA, PostgreSQL (AWS RDS)
Email	SendGrid Java SDK
Frontend	React 19, TypeScript, Material-UI v7
Containerisation	Docker, Docker Compose
Registry	AWS ECR
Compute	AWS EC2 (t3.micro)
Frontend Hosting	AWS S3 + CloudFront
Infrastructure	Terraform
Monitoring	Spring Actuator, Prometheus, Grafana, Node Exporter
CI/CD	GitHub Actions
Testing	JUnit, Spring Boot Test, H2 (in-memory)

Features

Secure Authentication & Authorization OAuth2-based auth with Keycloak, JWT access control, and role separation (Admin/User).	User-Scoped Data Management Full CRUD for birthdays with strict per-user isolation; admin visibility across all users.
Automated Email Reminders Scheduled daily jobs trigger birthday reminder emails via the SendGrid API.	RESTful API Design Clean REST endpoints with clear separation of controller, service, and persistence layers.
CI/CD Pipeline GitHub Actions for automated testing and zero-downtime deployment to AWS EC2 via ECR.	Infrastructure as Code Full AWS infra provisioned with Terraform: VPC, EC2, RDS, CloudFront, S3, and ECR.
Observability Stack Prometheus scrapes Actuator and Node Exporter every 15s; Grafana provides dashboards and alerts.	Containerised Deployment Docker Compose orchestrates API, Keycloak, Prometheus, Grafana, Node Exporter on one EC2 instance.

Architecture



Future Improvements

- * Implement API rate limiting for enhanced security
- * Add birthday notification preferences (lead time, channel)

Links

Backend (Spring Boot API): <https://github.com/keshav1207/RememberBirthdays>

Frontend (React + TypeScript): <https://github.com/keshav1207/RememberBirthdaysFrontend>

Live Demo: <https://dnvj8xyv0elqt.cloudfront.net>

Video Demo: <https://youtu.be/NufGv6MUI7s>